

EDUCATION	University of Pittsburgh Ph.D. Intelligent Systems Birla Institute of Technology and Science, Pilani M.E. Software Systems Thesis : Hyperspectral guided Image Dehazing B.E. (Hons.) Electrical & Electronics Thesis : Deep Learning for Biometrics	2021 - present <i>GPA : 3.7/4.0</i> 2018 - 20 2013 - 18
SELECTED PUBLICATIONS	Hybrid novice-AI system achieves expert-level performance in intraoperative ischemia detection N Murali, AI Mina, H Sinha , JW Anderson, Y Raka, H-C Chang, HK Amiri, PD Thirumala, K Batmanghelich, S Visweswaran <i>Under Review</i> spatiAlytica: Viewer-Grounded Multimodal Agentic System for Interactive Spatial Omics Analysis A Das, K Zhang, J Song, M Han, A Chen, W Meng, H Galloway, PY Chen, S Jo, Z Liu, MM Hasib, A Officer, H Sinha , YC Chiu, SJ Gao, L Li, Y Huang <i>Under Review</i> Spatial Single-Cell Atlas Reveals KSHV-Driven Broad Cellular Reprogramming, Progenitor Expansion, Immune and Vascular Remodeling in Kaposi's Sarcoma W Meng, A Das, H Sinha , R Naous, PM Bracci, M McGrath, Y Huang, SJ Gao <i>Under Review; bioRxiv preprint, 2025</i> Unifying AI-assisted scientific discovery around exploration, hypothesis generation, and testing MM Hasib, S Jo, H Sinha , et al. <i>In Patterns, 101660, 2026 (Review)</i> STiLE: Automated Tissue Microarray Dearraying for Spatial Transcriptomics H Sinha , A Das, YC Chiu, SJ Gao, Y Huang <i>In Bioinformatics, 2026</i> Solving the Pervasive Problem of Protocol Non-Compliance in MRI using an Open-Source tool <i>mrQA</i> H Sinha and PR Raamana <i>In Neuroinformatics, 2024</i> MRdataset: A unified and user-friendly interface to medical imaging datasets H Sinha and PR Raamana <i>In Journal of Open Source Software, 9(95), 6269, 2024</i> Towards Shape-regularized Learning for Mitigating Texture Bias in CNNs H Sinha and A Kovashka <i>In ACM International Conference on Multimedia Retrieval (ICMR 2023) Oral Session 🎤</i> Cycle Consistent Embedding of 3D Brains with Auto-Encoding Generative Adversarial Networks S Xing, H Sinha , and SJ Hwang <i>In International Conference on Medical Imaging with Deep Learning (MIDL 2021)</i>	

Domain-Aware Unsupervised Hyperspectral Reconstruction for Aerial Image Dehazing

A Mehta, [H Sinha](#), M Mandal and P Narang

In IEEE Winter Conference on Applications of Computer Vision (WACV 2021)

HIDeGAN: A Hyperspectral-guided Image Dehazing GAN

A Mehta, [H Sinha](#), P Narang and M Mandal

In IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW 2020)

Audio Classification using Braided Convolutional Neural Networks

[H Sinha](#), V Awasthi and Pawan K Ajmera

In IET Signal Processing, 2020

Text-Convolutional Neural Networks for Fake News Detection in Tweets.

[H Sinha](#), S Kalra and Y Sharma.

In Evolution in Computational Intelligence, Springer, 2020.

Upgrading Security and Protection in Ear Biometrics

[H Sinha](#) and PK Ajmera

In IET Biometrics, 2019

Convolutional Neural Network-based Human Identification using Outer Ear Images

[H Sinha](#), R Manekar, Y Sinha, and Pawan K Ajmera

In Soft Computing for Problem Solving, 2019

Studying the Role of Kinect as a Multi-Sensory Learning Platform for Children

[H Sinha](#), S Srivastava, and Y Sinha

In ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp 2018)

Interweaving Convolutions: An application to Audio Classification

[H Sinha](#) and PK Ajmera

In International Conference on Knowledge Discovery and Data Mining, DL Day Workshop (KDD 2018)

ABSTRACTS

A Spatial Single-Cell Atlas of Kaposi's Sarcoma: Insights into KSHV Tropism, Tumor Cell Origins, and Cellular Reprogramming in Tumor Progression, Angiogenesis, and Immune Evasion

W Meng, A Das, [H Sinha](#), M Han, R Naous, PM Bracci, MS McGrath, Y Huang, SJ Gao

In 27TH International Workshop on Kaposi's Sarcoma Herpesvirus (KSHV 2025)

Mapping the spatial biology of Kaposi Sarcoma (KS) in skin: Insights into tumor growth and cell behavior

A Das, W Meng, [H Sinha](#), SJ Gao, Y Huang

In 2024 CSBC / MetNet Annual Investigators Meeting (CSBC 2024)

Indiscriminate Adjustment for Confounders is worse than you think and What can be done about it

[H Sinha](#), and PR Raamana

In Alzheimer's Association International Conference (AAIC 2024)

Digging deeper into the pervasive problem of non-compliance in MR datasets

[H Sinha](#), and PR Raamana

In Organization for Human Brain Mapping Annual Meeting (OHBM 2024)| Oral Session 🎤

In ISMRM & ISMRT Annual Meeting & Exhibition (ISMRM 2024)

Solving the Pervasive Problem of Protocol Non-Compliance in MR imaging using an open-source tool mrQA

[H Sinha](#), and PR Raamana

In ISMRM & ISMRT Annual Meeting & Exhibition (ISMRM 2023) | Oral Session 
In Organization for Human Brain Mapping Annual Meeting (OHBM 2023)
In Alzheimer's Association International Conference (AAIC 2023)

Building bridges in MR research: unified interface to different MRI dataset formats with MRdataset
H Sinha, and PR Raamana

In INCF Neuroinformatics Assembly (INCF 2023) | Oral Session 

Bridging the Brain-Age Gap: Quantifying regional contributions using Shapley-Owen values
H Sinha, and PR Raamana

In Alzheimer's Association International Conference (AAIC 2023)

Garbage in, Garbage out : Protocol Compliance, Integrity and Quality Assurance of Neuroimaging Datasets

H Sinha, T Boonchalermvichien and PR Raamana

In INCF Neuroinformatics Assembly (INCF 2022)

Learning to Enhance Visual Quality via Hyperspectral Domain Mapping

H Sinha, A Mehta, M Mandal and P Narang

In AAAI Conference on Artificial Intelligence (AAAI 2021)

TEACHING

University of Pittsburgh

Introduction to Machine Learning

Spring 2024

Introduction to Computing for the Humanities

Fall 2023, Fall 2026

Discrete Structures for CS

Fall 2023, Fall 2026

Introduction to Computing for Scientists

Fall 2026

Johns Hopkins University

Reproducible Research

Summer 2015

Exploratory Data Analysis

Summer 2015

SOFTWARE

mrQA : Automatic protocol compliance checks on MR datasets



MRdataset : Unified interface to neuroimaging datasets



HONORS & AWARDS

ISMRM & ISMRT Annual Meeting & Exhibition Educational Stipend

2023, 2024

Intelligent Systems Program Fellowship

2021-2023

UbiComp 2018 Travel Grant

Oct 2018

Microsoft Code.Fun.Do. Finalist Forum

Feb 2016 - Aug 2016

REVIEWER

International Conference of IEEE Engineering in Medicine and Biology and Society (EMBC) 2025

Transactions on Circuits and Systems for Video Technology (IEEE-TCSVT)

2024-25

Journal of Open Source Software (JOSS)

2024

International Conference on Biomedical and Health Informatics (IEEE-EMBS BHI)

2024-25

British Machine Vision Conference (BMVC)

2020

Winter Conference on Applications of Computer Vision (WACV)

2021-23

European Conference on Computer Vision (ECCV)

2022

IET Biometrics

2019-25

TECHNICAL SKILLS

Languages: PyTorch (Geometric, Lightning), R, Keras, Python, SQL, Java, C/C++

Courses: Causal Inference, Machine Learning, Computer Vision, Object Oriented Programming, Multimodal Neuroimaging, Data Mining, Information Retrieval, Artificial Intelligence, Probability and Statistics

MOOCs: Data Science Specialization